

Ending a Resuscitation

Section 1: Case Summary

Scenario Title:	
Keywords:	Death, hot debrief, breaking bad news
Brief Description of Case:	The medical aspects of this case are a straight-forward cardiac arrest where the team must recognize futility and make the decision to stop resuscitation efforts. The primary goal is experiencing making a decision on calling a code and managing the impacts of a patient's death. This provides additional optional opportunities: <i>Optional (junior or senior)</i> – breaking bad news to family member <i>Optional (senior)</i> – simulated hot debrief of the case with the team Other goals could also be scaffolded onto this scenario as deemed appropriate by the simulation instructor.

Goals and Objectives	
Educational Goal:	1. Recognize futility in resuscitation and when to terminate resuscitation 2. Affective experience of decision for termination and of a patient death <i>Optionally</i> 3. Demonstrate effective and empathetic breaking of bad news 4. Ability to lead a hot debrief of an emotionally charged case
Objectives: (Medical and CRM)	CRM: 1. Effective leadership and teamwork of a standard ACLS case Medical: 1. Demonstrate standard ACLS management of a cardiac arrest 2. Recognize futility in resuscitation and demonstrate ability to appropriately terminate resuscitation Other: 1. Manage the emotional and affective experience of a decision for termination and of a patient death 2. Demonstrate effective and empathetic breaking of bad news (<i>optional</i>) 3. Facilitate a team hot debrief (<i>optional</i>)
EPAs Assessed:	TD 1: Recognizing the unstable/critically ill patient, mobilizing the health care team and supervisor, and initiating basic life support F1: Initiating and assisting in resuscitation of critically ill patients C1: Resuscitating and coordinating care for critically ill patients C3: Providing airway management and ventilation

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Learners, Setting and Personnel					
Target Learners:	☒ Junior Learners		☒ Senior Learners		☐ Staff
	☒ Physicians		☐ Nurses	☐ RTs	☐ Inter-professional
	☐ Other Learners:				
Location:	☒ Sim Lab		☐ In Situ		☐ Other:
Recommended Number of Facilitators:	Instructors: 1				
	Sim actors: 1				
	Sim Techs: 1				

Scenario Development	
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Last Revision Date:	-
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Section 2A: Initial Patient Information

A. Patient Chart					
Patient Name: Jane Doe			Age: 60	Gender: F	Weight: 100 kg
Presenting complaint: Cardiac arrest					
Temp: 35.8	HR: 0	BP: 0	RR: 0	O ₂ Sat: n/a	FiO ₂ : 100%
Cap glucose: 5.5			GCS: 3 (E1 V1 M1)		
Triage note: Out of hospital cardiac arrest. EMS called for chest pain. Patient collapsed on EMS arrival, witnessed arrest, CPR started. Total transport time 30 minutes. Shock x1 for ventricular fibrillation. Now in asystole.					
Allergies: None					
Past Medical History: CAD History MI x2, stents Hypertension Dyslipidemia			Current Medications: ASA Ramipril Metoprolol Rosuvastatin		

Section 2B: Extra Patient Information

A. Further History	
Additional history is given from RN who received EMS handover. Information should be readily and easily available – it should be clear what is going on. At 1 st pulse and rhythm checks by EMS was in VF → shocked At 2 nd and 3 rd pulse and rhythm check by EMS was in asystole → running ACLS in asystole since. Has received epinephrine x2 doses. Did not receive amiodarone (was in asystole at 2 nd and 3 rd rhythm check). IV access x1 18g AC. Total downtime thus far has been 30 minutes. Goals of care previously documented as full resuscitation Family is on their way in. Baseline level of function appears to be independent / high functioning.	
B. Physical Exam	
Cardio: Cardiac arrest, no other findings	Neuro: Pupils 3mm, fixed
Resp: Apnea, no other findings	Head & Neck: No findings
Abdo: No findings	MSK/skin: Cool and mottled extremities.
Other: Palpable pulse with CPR. If an arterial line is placed, diastolic BP with CPR is 20. Quantitative ETCO ₂ is 8. POCUS shows absence of organized cardiac contractility	



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Section 3: Technical Requirements/Room Vision

A. Patient
☒ Mannequin - adult (<i>specify type and whether infant/child/adult</i>)
☐ Standardized Patient
☐ Task Trainer
☐ Hybrid
B. Special Equipment Required
Crash cart + ACLS equipment Airway equipment Arterial line equipment Point-of-care ultrasound IV equipment Intra-osseous equipment
C. Required Medications
ACLS medications – epinephrine, amiodarone Intubation medications – ketamine, propofol, rocuronium, succinylcholine IV fluid
D. Moulage
None needed
E. Monitors at Case Onset
☒ Patient on monitor with vitals displayed ☐ Patient not yet on monitor
F. Patient Reactions and Exam
No reactions. Patient is unresponsive. No respiratory efforts. Eyes closed.

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Section 4: Sim Actors and Standardized Patients

Sim Actors and Standardized Patient Roles and Scripts	
RN	Assists with ACLS – monitors, medications, etc. Provides clarification to history and exam if needed (it should be made obvious to the team what is going on): - Provide details from EMS handover (see below) - Family is on their way in - Goals of care previously documented as full resuscitation - Sounds like baseline level of function has been independent / high functioning EMS handover that was given to RN: - At 1st pulse and rhythm checks by EMS was in VF → shocked - At 2nd and 3rd pulse and rhythm check by EMS was in asystole → running ACLS in asystole since. - Has received epinephrine x2 doses. Did not receive amiodarone (was in asystole at 2nd and 3rd rhythm check). - IV access x1 18g AC. - Total downtime thus far has been 30 minutes. Prompt if no termination of resuscitation after >15 min, RN prompts, “How long are we going to do this for? Should we just call this?”. If they are not getting the point, the RN can additionally prompt: “I’m uncomfortable with how long we have run this for.” or “They can’t have much brain left anymore... I can’t imagine them having any quality of life after this.”
RN during debrief <i>(optional)</i>	If choosing to do a hot debrief, the RN will be the actor (i.e. playing their role) during this debrief as well. The facilitator may also join the simulated hot debrief as a senior nurse, charge nurse, care facilitator, or social worker. The RN will initially present him/herself as unaffected by the death (“we see this all the time, nothing was special about this”). They should initially be reserved and not very open about sharing their opinions or feelings. As the discussion continues should develop more self-doubt (“I just wonder if we could have done more”). If the participant leading the debrief is empathetic, genuine, and seems effective, then the nurse will respond positively to this.
Family member <i>(optional)</i>	The family (son or daughter, and/or husband) will arrive within 1 minute of termination of resuscitation, or after 15 minutes of case time if the team has not made this decision. Goals of care: “Do everything you can, unless you think she won’t be able to eventually recover in a meaningful way”. If questioned further, she would not accept substantial loss of function, and would not want “life support for a prolonged time” but would accept it if chance for recovery is reasonable.



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The family member lives separately from their parent, but in the same city. They visit about once per week. Their parents are both independent, active around the house, visit friends outside of the house, do their own groceries/banking/laundry, etc.
High functioning baseline.

Further backstory encouraged (at the discretion of the instructors): e.g. they enjoy camping and tail-gating at local college football games. They were planning a road trip soon.

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Section 5: Scenario Progression

Scenario States, Modifiers and Triggers				
Patient State/Vitals	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State		Facilitator Notes
1. Baseline State Rhythm: Asystole HR: 0 BP: 0 RR: 0 O ₂ SAT: not detected T: 35.8°C GCS: 3	Patient is on stretcher with CPR ongoing Not on monitor	<u>Expected Learner Actions</u> ☐ RN/EMS handover ☐ Attach monitors / crash cart ☐ Rapidly identify asystole ☐ Get second IV ☐ Continue ACLS management ☐ Intubation ☐ Epinephrine on next cycle ☐ Labs ☐ Try to obtain goals of care ☐ Voice futility (when appropriate) and terminate resuscitation	<u>Modifiers</u> - if arterial line → BP 30/20 during CPR only - if ETCO ₂ → display CO ₂ = 8 <u>Triggers</u> - If >15 min passes with resident calling for termination → RN prompts for termination <i>Depending on case goals, after TOR the case can end one of 3 ways:</i> 1. Simple death (end of case) 2. Family arrives → go to break bad news stage 3. TOR + RN asks for debrief → go to hot debrief stage	If POCUS done - asystole → no cardiac activity Case ends in one of three ways: 1. TOR → nothing further 2. Family arrives → need to break bad news 3. TOR + RN asks for debrief
2. Break Bad News <i>(optional)</i>	Family arrives after TOR or if >15 min pass	<u>Expected Learner Actions</u> ☐ Introduce self / team ☐ Use family member names ☐ Appropriate body language ☐ Reasonable approach to breaking news (e.g., SPIKES)	<u>Modifiers</u> - Turn off monitors if resident asks for them to be turned off. <u>Triggers</u> - At natural conclusion of conversation or if >10 min have passed, leader is called overhead to the resus bay (end of case)	Breaking bad news should occur immediately following the case (prior to debrief of the simulation) and involve the leader and any team members (max 4). A second room is optional, but a space away from the mannequin with chairs is ideal.
3. Hot Debrief <i>(optional)</i>	RN asks the leader for a team debrief	<u>Expected Learner Actions</u> ☐ Introduce the debrief and provide a structure ☐ Establish psychological safety	<u>Modifiers</u> - Turn off monitors if resident asks for them to be turned off.	Hot debrief of the team should occur immediately following the case (prior to debrief of the



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		☐ Acknowledge emotions ☐ Discuss team successes ☐ Discuss areas for improvement ☐ Provide some form of closure to the debrief and incident	<u>Triggers</u> - At natural conclusion of conversation or if >10 min have passed, leader is called overhead to the resus bay (end of case)	simulation) and involve the leader and any team members (max 4). A second room is optional, but at least a small space in the room away from the mannequin should be used.
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Appendix A: Laboratory Results

<u>CBC</u> WBC 14 Hgb 130 Plt 367 <u>Lytes</u> Na 137 K 4.7 Cl 113 HCO ₃ 14 AG 10 Urea 6 Cr 80 Glucose 5.5 <u>Extended Lytes</u> Ca 2.21 Mg 1.10 PO ₄ 0.94 Albumin 35 <u>VBG</u> pH 7.05 pCO ₂ 50 pO ₂ 65 HCO ₃ 16 Lactate 5	<u>Cardiac/Coags</u> Trop < 14 D-dimer 200 INR 1.2 aPTT 40 <u>Biliary</u> AST 30 ALT 20 GGT 30 ALP 15 Bili 9 Lipase 14 <u>Tox</u> EtOH negative ASA negative Tylenol negative <u>Other</u> B-HCG negative
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Appendix B: ECGs, X-rays, Ultrasounds and Pictures

CXR - intubated



Source: <https://emcow.files.wordpress.com/2012/11/normal-intubation2.jpg>

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Appendix C: Facilitator Cheat Sheet & Debriefing Tips

Patient death

There is some controversy over whether to allow a simulated patient to die when death is not a predefined learning objective¹. When it is a goal of the session, simulation can provide a safe yet realistic environment for experiencing death of a patient and exploring the emotional impacts.

The debrief can go in many different ways and should allow proper exploration of learner reactions and emotions. At a minimum, debrief should cover the affective impact of the scenario and physician/learner wellness.

Discussion topics:

1. Affective objectives
 - a. Emotional impact of death
 - b. Explore team elements in shared responsibility and allowing team input. Does the team feel comfortable with the decision made? Should other avenues be explored or acknowledged to allow for psychological safety of team members?
 - c. Identity of physician role in determining death – death, futility of resuscitation, and expected prognosis are often determined by a physician, with great weight on these decisions.
2. Physician / personal wellness
 - a. Normalize the importance of wellness and personal impact death can have on us as people
 - b. Acknowledge the importance of self-care, personal wellness strategies, reflecting on emotional impacts
 - c. Talk about reaching out for help, local resources for help, etc.
3. Cognitive objectives
 - a. What are the indicators in general and in this case that suggest futility in resuscitation, and appropriateness of termination of efforts? E.g.,
 - i. Poor prognostic signs
 - Downtime >20 minutes
 - ETCO₂ <10 at 20 minutes resuscitation
 - Asystole is poor prognostic sign
 - Absence of cardiac activity or signs of life
 - ii. Good prognostic signs
 - Witnessed arrest and immediate CPR are good prognostic signs
 - Early shock for VF
 - b. From an ACLS perspective, what are the appropriate therapies, and what are their relative impacts on likelihood of successful resuscitation? e.g.,
 - i. Early VF has NNT 5 for survival
 - ii. PARAMEDIC-2 trial - NNT 112 for survival with epinephrine, but with poor neurologic outcomes
 - iii. ALPS trial - no difference in survival to discharge between amiodarone, lidocaine, or placebo



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Breaking bad news

Breaking bad news is a crucial skill for ED physicians. There are many approaches, but whatever the approach it is important to have a consistent and effective approach one can fall back on in a pinch or when time is pressured.

One example is the SPIKES approach² to breaking bad news, adapted from oncology literature:

- S - Create a SETTING conducive to the discussion.
- P - Learn about the patient's PERCEPTION of their illness.
- I - Obtain the patient's INVITATION to discuss it.
- K - Share your KNOWLEDGE about the diagnosis, etiology, prognosis, and management.
- E - Explore their EMOTIONS and EMPATHIZE to validate patient feelings.
- S - SUMMARIZE and develop and shared plan and STRATEGY.

Hot debrief

A "hot" debrief, as opposed to a "cold" debrief, happens immediately after the incident and is much more burdened with emotional impact. There are often competing time demands as well. Team members often feel there would be benefit to immediate debriefs, though it is uncommon to seek them out. They theoretically can give the team an opportunity to process emotional impacts and give insight into improving future performance, though evidence is still emerging.

STOP5 is one approach to hot debrief⁴:

- S - Summarize the case
- T - Think about what went well
- O - Options for improvement
- P - Points to action or responsibilities
- 5 - STOP for 5 minutes to debrief

Another general approach³ involves,

1. Establish the framework for the debrief
 - Provide an invitation, in a non-judgmental manner, to participate in a debrief
 - Time frame (5 -10 minutes)
 - Provide opportunity to return to other clinical work if very pressing
2. Basic assumption
 - Establish a safe learning environment for team members
3. Discuss successes (what went well?)
 - Leadership
 - Communication
 - Team dynamic
4. Discuss challenges (what could be improved upon?)
 - PPE
 - Equipment failure
 - Delay to therapy
5. Assess prevention of incident (could this have been prevented?)
 - What led up to the event?
6. Consider if a cold debrief is needed at a future time for this event
7. Summary
 - Thank all members for participating
 - Make yourself available for individual discussion immediately after and in the future

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References

1. Corvetto MA, Taekman JM. To die or not to die? A review of simulated death. *Simul Healthc*. 2013;8(1):8-12. doi:10.1097/SIH.0b013e3182689aff
2. Buckman R. Breaking bad news: the S-P-I-K-E-S strategy. *Community Oncology*, 2005; 2(2). DOI: 10.1016/S1548-5315(11)70867-1.
3. Bhatnagar P. Debrief while it's HOT. CanadiEM. <https://canadiem.org/debrief-while-its-hot/>. Published April 23, 2020. Accessed January 17, 2021.
4. Walker CA, McGregor L, Taylor C, Robinson S. STOP5: a hot debrief model for resuscitation cases in the emergency department. *Clin Exp Emerg Med*. 2020;7(4):259-266. doi:10.15441/ceem.19.086
5. Couper K, Perkins GD. Debriefing after resuscitation. *Curr Opin Crit Care*. 2013;19(3):188-194. doi:10.1097/MCC.0b013e32835f58aa

